In the beginning, there was the singularity.

Well, maybe. A singularity is what most people think of when they think of the Big Bang: an infinitely dense point from which everything in the universe exploded outward. Only, a singularity doesn't have to be a point—it could just be an infinitely dense state of an infinitely large universe. And, as discussed above, there's no explosion per se, since an explosion implies an expansion *into something*, rather than an expansion *of everything*. The idea that everything started with a singularity comes from observing the current expansion of the universe, applying Einstein's equations of gravity, and extrapolating backward. But that singularity might never have happened. What most physicists do think happened, a fraction of a second after whatever was the true "beginning," was a dramatic super-expansion that effectively erased all trace of whatever went on before it. So the singularity is one hypothesis for what might have started everything off, but we can't really be sure.

There's also the question of what was "before" the singularity. That question might be, depending on who you ask, incoherent nonsense (because the singularity was the beginning of time as well as space, so it had no "before"), or one of the most crucial questions in cosmology (because the singularity might have been the end point of a previous phase of a cyclic universe: one that goes from Big Bang to Big Crunch and back again forever). We'll talk about the latter possibility in Chapter 7, but in the meantime, there's not much to say about the singularity other than that it might have happened. Even if we did trust ourselves to dial back expansion all the way to that point, a singularity represents a state of matter and energy so extreme that nothing we currently know about physics can describe it.

To a physicist, a singularity is pathological. It's a place in the equations where some quantity that is normally well behaved (like the density of matter) goes to infinity, at which point there is no longer any way to calculate things that makes any sense. Most of the time, when you encounter a singularity, it is telling you that something has gone wrong in your calculations and you need to go back to the drawing board. Finding a singularity in your theory is like having your satellite navigation direct you to the edge of a lake, then instruct you to disassemble your car, reassemble it as a boat, and paddle your new car-boat to the other side. Maybe this